

Employee CV

MARTIN GRELLMANN

Senior Engineer | Legacy Modernization | AI-Assisted Code Analysis | ABAP & PHP

Location: Goiás, Brazil (America/Sao_Paulo, BRT, UTC-03:00, no DST)

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German citizen with unrestricted work and residence permit in Brazil.

Languages: German — Native

English — Professional working proficiency

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Professional Summary

Senior engineer for legacy modernization, with a working method that combines AI-assisted code analysis with strict manual verification.

I take over systems nobody understands any more: undocumented ABAP applications from the nineties, heavily customized Drupal 7 multisites, grown PHP backends. AI helps me read and summarize large amounts of unknown code quickly; I verify every finding against the running system before anything is changed.

This is not a tooling experiment. It is my daily work in two productive environments right now, one SAP and one PHP, and it shortens the analysis phase that used to make legacy projects unaffordable.

Core Strengths

- AI-assisted analysis of undocumented code, with a verification step for every finding before it is used.
- Reconstruction of business rules from legacy code and writing the documentation that never existed.
- Incremental modernization in productive systems: no big-bang, each step can go live.
- Two languages, one method: object-oriented ABAP in SAP, PHP and JavaScript on the web side.
- Cleaning up what usually comes with legacy: leaked credentials, missing environment separation, undocumented deployments.
- Honest about the limits of AI in this work, and clear about where a human has to decide.

How I Work on Legacy Systems

- Inventory first: what exists, what is really called, what is dead code.
- AI-assisted reading of the unknown code to get a fast first map of modules, dependencies and rules.
- Verification of every AI finding against the running system, the database and the people who use it. Nothing goes into a decision unverified.
- Documentation of the reconstructed rules before any rewrite, so the knowledge survives the project.
- Incremental rewrite with the safest slice first, each step releasable on its own.
- Handover: documentation, conventions and a working setup the customer team can continue with.

Professional Experience

Technical Lead / IT Specialist, Ferienhausvermittlung fincaferien Düsseldorf, Germany

03/2025 to Present

Technical lead for the rescue of a heavily customized Drupal 7 multisite platform with seven productive sites, an undocumented custom module base and a five-year maintenance backlog.

- Took over a Drupal 7 multisite system with no documentation, no environment separation and credentials committed in plain text, and brought it back under control.
- Used AI-assisted code analysis to read and summarize the undocumented custom modules, then verified every finding against the running system before acting on it.
- Introduced a clean settings split (versioned configuration, unversioned secrets), a redaction workflow for terminal output and pre-commit checks so credentials stop leaking.
- Separated development and production trees, moved the platform to cloud infrastructure and documented the previously undocumented deployment.
- Repaired the booking and availability processes and wrote the missing documentation as I went, so the knowledge no longer lives only in the code.

Technologies

Drupal 7, PHP, MySQL/MariaDB, Linux, Git, Drush, cloud infrastructure, AI-assisted code analysis, documentation.

Senior SAP Developer, intecsoft GmbH & Co. KG Dresden, Germany

01/2025 to Present

Senior SAP development role focused on modernization, integration and software quality in S/4HANA projects.

- Reverse-engineered an undocumented SAP Business Suite Dynpro application in IS-U and rewrote it in object-oriented ABAP, using AI-assisted code reading to speed up the analysis of the original logic.
- Reconstructed and documented the business rules that only existed in the code.
- Rebuilt backend API interfaces in an S/4HANA project and kept productive services stable during hypercare in FI.
- Worked with business teams and solution managers to confirm the reconstructed behaviour before replacing the old code.

Technologies

ABAP Objects, ABAP SQL, CDS Views, OData, SAP BTP, REST, Fiori/UI5, S/4HANA, Clean Core.

SAP Technical Consultant, BA Business Advice GmbH Oldenburg, Germany

02/2022 to 12/2024

SAP consulting and development with a focus on migration and refactoring of grown, poorly documented custom applications.

- Migrated and refactored twenty-year-old FI and PM custom applications from SAP R/3 to S/4HANA, starting from code with almost no documentation.
- Reconstructed business rules from grown procedural code and rewrote them as object-oriented ABAP.
- Contributed to a data migration from a self-developed C++ ERP system to S/4HANA.
- Worked in the cloud expert team for S/4HANA Cloud Public Edition, BTP and ABAP Clean Core.
- Delivered in the project areas PPDS, QM, LE and PM.

Technologies

ABAP, ABAP Objects, ABAP SQL, CDS Views, OData, SE80, Eclipse, S/4HANA, SAP BTP, Clean Core.

Software Developer and Linux Administrator, Sengicon GmbH
Leipzig, Germany

02/2021 to 01/2022

Full-stack development with a focus on turning an outdated PHP application into a maintainable REST backend with a modern frontend.

- Rebuilt an outdated PHP web application into a PHP REST backend with a newly built React frontend, step by step and without stopping daily operations.
- Refactored grown PHP code and reshaped database schemas that had drifted from the data model.
- Refactored existing React components alongside the backend work.
- Administered Linux VMs and LAMP stacks for development and production.

Technologies

React, JavaScript, PHP, REST, PHPUnit, Jest, Linux, Apache, MySQL, MariaDB, PostgreSQL, Composer.

Webmaster / Web Developer, Ferienhaus-Vermittlung fincaferien.de GmbH
Bockenem, Germany

04/2002 to 12/2019

Long-term responsibility for the web platform that I am modernizing again today, which gives me first-hand knowledge of how the legacy system grew. Started in 04/2002 as a part-time role of about ten hours per week alongside my apprenticeship as an Industrial Clerk, with full technical responsibility from 04/2004.

- Developed and maintained the Drupal 7 multisite platform and its custom modules from 2015 onwards.
- Introduced Git and Jira into a workflow that had none before.
- Increased organic traffic through technical SEO, performance and usability work.
- Managed Linux, Samba and Windows network environments.

Technologies

PHP, Drupal 7, HTML, CSS, JavaScript, MySQL, Git, Jira, Linux, Samba, SEO/SEA.

Selected Projects

Migration of custom applications from R/3 to S/4HANA

04/2024 to 09/2024

- Refactored and migrated FI and PM custom applications to modern ABAP.
- Modernized applications that had grown over twenty years.

Data migration from a C++ ERP system to S/4HANA

03/2022 to 05/2022

- Supported the migration of a self-developed ERP system to S/4HANA for a wire processing customer.

Bloom optimizer, custom ABAP application (PPDS)

05/2022 to 06/2024

- Developed a bloom optimizer for a steel mill that plans rolling processes by steel grade, loading location, minimum remaining ends and technical constraints.
- Implemented in Eclipse and SE80 and integrated into PPDS.

Technical Skills

Legacy analysis	Reverse engineering of undocumented applications, dead code detection, dependency mapping, reconstruction and documentation of business rules, AI-assisted code reading with manual verification.
SAP	ABAP, ABAP Objects, ABAP SQL, Dynpro, CDS Views, OData, RAP, SAP BTP, S/4HANA, Clean Core, SE80, ABAP Development Tools for Eclipse.
Web	PHP, Drupal 7, Composer, JavaScript, TypeScript, React, Node.js, REST APIs, PHPUnit, Jest.
Data	MySQL, MariaDB, PostgreSQL, schema archaeology, migration of grown data models.
Operations	Linux administration, Apache, LAMP, Drush, cloud infrastructure, Git, environment separation, secret handling and credential rotation.
Method	Clean Code, SOLID, refactoring patterns, architecture documentation, incremental rewrite, Scrum (PSM I), Jira, OpenProject.

Education and Training

Software Architecture, Software Academy by intecsoft, Dresden, Germany iSAQB CPSA-F Foundation Level training (online, 14–17 July 2025), certificate of attendance; exam planned for Q4/2026.	07/2025
Full-Stack Web Developer, Indiesoft GmbH, Leipzig, Germany Full-Stack Web Developer certification (HTML, CSS, JavaScript, PHP, React, databases).	2020 to 2021
Agile Project Management, Indiesoft GmbH, Leipzig, Germany Professional Scrum Master I (PSM I), certified by Scrum.org in June 2020.	05/2020 to 06/2020
Industrial Clerk, DAA GmbH, Hildesheim, Germany Apprenticeship as Industrial Clerk (Industriekaufmann).	2002 to 2004

Certificates and References

- Reference letter BA Business Advice GmbH (31.12.2024): SAP ABAP, MII, UI5 Freestyle, Fiori Elements, cloud expert team, ERP to S/4HANA migration, refactoring, ABAP Clean Core. Overall rating: very good.
- Reference letter Sengicon GmbH (21.12.2021): PHP backend development and refactoring, React frontend, database modelling, Linux administration. Overall rating: very good.
- Reference letter fincaferien.de GmbH (31.12.2019): web development, Drupal 7, SEO/SEM, usability, IT and network. Overall rating: very good. Position ended for operational reasons.
- Certificates: Professional Scrum Master I (Scrum.org, 2020), Full-Stack Web Developer (2021), Agile Project Management (2021), Object-Oriented PHP (2020).